

LESSON 1.6

WRITING AND SOLVING ONE-STEP EQUATIONS IN REAL-WORLD CONTEXTS

LESSON INTRODUCTION

MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.

MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.

LEARNING TARGETS

- I can write and solve one-step addition, subtraction, multiplication, and division equations that represent a real-world context.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

ESSENTIAL QUESTIONS

- How can I write a one-step algebraic equation from a real-world situation?

LESSON NARRATIVE

In this lesson students write and solve algebraic equations to solve real-world problems. Students work collaboratively frequently throughout the lesson, so they have multiple opportunities to discuss the problems presented. First, students engage in a collaborative activity where they examine equations and decide which problems they represent, focusing in on the appropriate operations. Then, students complete a Guided Practice in writing and solving equations that model real-world scenarios. Students write their own problems that could be modeled by the given equations and compare their problems with a partner.

COMPONENT SUMMARY

1.6.1 WARM-UP (~5 MIN.)

Mathematical thinking and reasoning

1.6.3 GUIDED PRACTICE (15 MIN.)

Writing one-step equations from situations

1.6.5 WRAP-UP (~5 MIN.)

Ticket Out the Door on writing and solving one-step equations

1.6.2 COLLABORATION (10 MIN.)

Determining an equation for a real-world situation

1.6.4 COLLABORATION (10 MIN.)

Determining a situation for a one-step equation

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Inverse operation
- Equation
- Revenue

MATERIALS

- (optional)

ADDITIONAL PREPARATION

- Consider providing select students with graphic organizers for common key words (i.e., split indicated division) to help students decode word problems.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students frequently confuse problems that call for multiplication and those that call for division.
- Students may get stuck understanding what operation is shown in division equations (i.e., $\frac{x}{5} = 3$) because they will see it as a fraction and not see the division.
- Students may struggle what operation is being used.

THINGS TO CONSIDER

- Encourage flexibility in mathematical thinking and reasoning. Use language and phrases like, “Your process matters more than your product” or “Contemplate before calculating” to encourage an emphasis on the operations and analysis rather than on the calculations.
- Consider placing students as partners based on observations, student self-reflections, and Tickets Out the Door.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

In 1.6.3 Guided Practice, have students engage in a Notice and Wonder. Prompt students to think about the following questions: What do you notice about the division equation and multiplication equation? What do you wonder? Can inverse operations always be used to write an equation?

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

In this lesson, students are able to connect one-step algebraic equation to real-world situations. Students are provided equations and need to determine a real-world situation where that equation could be used to solve a problem.

DIFFERENTIATE INSTRUCTION

The On-Ramp to 7th Grade Accelerated Math video and practice tool offers a variety of previously learned skills instruction and practice to support all students. Consider having students preview this lesson by watching the On-Ramp video or have the video available to watch for students to opt-in throughout the lesson.

- Verifying Solutions of Algebraic Equations

1.6.1 WARM-UP: WOULD YOU RATHER?

DIFFERENTIATE INSTRUCTION

Consider reviewing the word “revenue” to give all students entry into this Warm-Up.

TEACHER MOVES

You are not looking for a correct answer here, but rather that students are using mathematical thinking and reasoning to explain their choice. Begin prompting this mindset during the Warm-Up to set up students for strong math thinking and reasoning during this lesson.



1.6.1 Warm-Up: Would You Rather?

The number of rooms and rate per night at two different Florida hotels are shown.

Downtown Miami Hotel	Luxury Key West Resort
 527 rooms at \$139 per night	 40 cottages at \$1,049 per night

- From which business would you rather earn the revenue?

Answers will vary.

If all rooms are booked at both businesses, I would rather earn revenue from the Downtown Miami Hotel. The Downtown Miami Hotel has more rooms and would bring in more money per night.

- Explain your reasoning using mathematics.

Answers will vary.

$527 \times 139 = \$73,253$ per night (Downtown Miami Hotel)

$40 \times 1049 = \$41,960$ per night (Luxury Key West Resort)



1.6.2 Collaboration: Writing and Solving Equations in Real-World Context

Work with your partner to complete the following.

- 1. Diandra’s cat weighs 6 pounds fewer than Kim’s puppy. Kim’s puppy weighs 17 pounds.
 - a. Select all of the equations that could be used to find the weight of Diandra’s cat.
 - ☒ $x + 6 = 17$
 - ☐ $x - 6 = 17$
 - ☐ $x = 17 + 6$
 - ☒ $x = 17 - 6$
 - ☐ $x - 17 = 6$
 - ☐ $x + 17 = 6$
 - b. What does the variable, x , represent in this situation?
 x represents the weight of Diandra’s cat.
 - c. What is the weight of Diandra’s cat?
Diandra’s cat weighs 11 pounds.
- 2. Four equations are shown.

$6x = 42$	$42x = 6$
$\frac{x}{42} = 6$	$\frac{x}{6} = 42$

- a. Discuss with your partner which of the equations correctly represents each situation below, where x represents the unknown quantity.

Situation 1	Situation 2
At a restaurant, 6 friends decided to split the bill equally. If each person paid \$42, what was the total bill?	A small theater has a total of 42 seats. If the theater has 6 rows of seats with the same number of seats in each row, how many seats are in each row?

1.6.2 COLLABORATION: WRITING AND SOLVING EQUATIONS IN REAL-WORLD CONTEXT

TEACHER MOVES

Notice question 1a is a “select all,” so prompt students to continue looking after they have found one equation that could be used.

ENRICHMENT

How can you determine what the variable represents in the situation?

TEACHER MOVES

Prompt students to focus on the appropriate operations when determining the matching equation. Students can highlight or underline key words in the situation that indicate which operation(s) should be used.

TEACHER MOVES

Notice that either division equation is correct for Situation 2. Both yield the same solution and both work for x to represent the total number of seats in the theater.
This is not the case for the multiplication equation. Only $6x=42$ is the correct equation for the situation.

1.6.2 COLLABORATION: WRITING AND SOLVING EQUATIONS IN REAL-WORLD CONTEXT (CONTINUED)

DIFFERENTIATE INSTRUCTION

If students are struggling to determine which equation will work for the situation, prompt students to first determine how they would solve, and use that to guide them toward the equation.

ENRICHMENT

MA.K12.MTR.2.1: Multiple Representations

Why do both division representations work for Situation 2? Does the variable represent two different things in each equation? How would you describe each?

b. Write the matching equation in the space provided.

Situation 1	Situation 2
Equation:	Equation:
$\frac{x}{6} = 42$ or $\frac{x}{42} = 6$	$6x = 42$

c. What does x represent in each equation?

Situation 1	Situation 2
In this situation, the quantity x represents ...	In this situation, the quantity x represents ...
x represents the total cost of the bill.	x represents the number of seats in each row.



1.6.3 Guided Practice: Writing and Solving Real-World Problems with One-Step Equations

For each situation, write an equation that can be used to determine the unknown quantity. Then, solve the equation. Be sure to check your solution.

- Carl, a student at R.J. Murray Middle School in St. Augustine, is preparing for a trip to Universal Studios with his mom and dad. An adult ticket is \$164. A ticket for a child under 18 is \$5 less than an adult ticket.
 - Write an algebraic equation to represent the relationship between the cost of a child's ticket and the cost of an adult ticket in this situation.
 $x + 5 = 164$
 - What does the variable in your equation represent?
The variable x represents the cost of the child's ticket.
 - Solve the equation.

$$\begin{array}{r} x + 5 = 164 \\ -5 \quad -5 \\ \hline x = 159 \end{array}$$
 - What was the cost of Carl's ticket?
\$159.00
- Gabriela is a seamstress who owns her own business. One of her specializations is Quinceañera dresses. It takes 5 yards of fabric to make one dress.
 - Write a division equation and a multiplication equation that can be used to determine how many yards of fabric, d , are needed to make 16 Quinceañera dresses.

Division Equation	Multiplication Equation
$\frac{d}{5} = 16$	$16 \cdot 5 = d$

- Solve either of the equations for d . Use the other equation to verify your solution is correct.

$$\begin{array}{ll} 5 \times \frac{d}{5} = 16 \times 5 & 16 \cdot 5 = d \\ d = 80 & d = 80 \end{array}$$

1.6.3 GUIDED PRACTICE: WRITING AND SOLVING REAL-WORLD PROBLEMS WITH ONE-STEP EQUATIONS

TEACHER MOVES

Make decisions for how students engage with this activity based on your observations throughout Unit 1. Students may need a model or Think Aloud, additional support with a peer, or independent practice and review.

ENRICHMENT

MA.K12.MTR.6.1: Reasonableness

Is your answer reasonable, or does it make sense in this context? Why or why not?

TEACHER MOVES

Some students may become frustrated with having to write equations for problems that they can easily find the answer to or they write an equivalent form that shows the solution. Use question 2a to establish these equations as equivalent. If necessary, establish the form for which the equation should be written such as $x + p = q$, $x - p = q$, $px = q$, or $x/p = q$. Question 2a is purposely designed to show that there are equivalent equations.

1.6.4 COLLABORATION: WRITING REAL-WORLD SCENARIOS TO REPRESENT ALGEBRAIC EQUATIONS

DIFFERENTIATE INSTRUCTION

Universal Design for Learning

Notice the open-ended nature of this activity. Students are encouraged to make meaningful connections to their own personal experiences. Emphasize students as experts of their own experiences as they collaborate, positioning each student as the driver of their co-teaching and co-learning in this activity.

TEACHER MOVES

For students struggling to write a real-world scenario, prompt them to solve to find the solution, then choose what that solution could represent.

For example, students may struggle to create a meaning for $21y = 105$, but students can produce a meaning for the number 5, then deciding on what relationship that 5 something could have with the coefficient 21.

ENRICHMENT

How can we identify what the variable should represent in a given problem?



1.6.4 Collaboration: Writing Real-World Scenarios to Represent Algebraic Equations

- For each equation, create a real-world problem that the algebraic equation could represent.
Answers will vary.

Equation	Your Real-World Scenario	What does the variable represent in your scenario?
$r - 10 = 12$	<i>Cayla loves playing basketball. This weekend she made 10 more shots during her game. Last weekend she made 12 shots. How many shots did Cayla make this weekend?</i>	<i>The number of shots Cayla made this weekend.</i>
$21y = 105$	<i>Maurice bought burgers for all of his friends from an expensive restaurant. Each burger was \$21, and his bill was \$105. How many burgers did he buy?</i>	<i>Number of burgers Maurice purchased.</i>
$\frac{a}{10} = 3$	<i>Shaniqua bought treats for her class party. She divided the treats evenly among her 10 classmates. Every student received 3 treats. How many treats did she purchase?</i>	<i>The total amount of treats Shaniqua purchased.</i>
$17 + s = 26$	<i>Asher had \$17. He did chores and made some money. He now has \$26. How much did he earn from completing his chores?</i>	<i>The money he earned by doing chores.</i>

2. Pick two equations from the table and compare your scenarios and solutions with your partner. Use them to complete the table below.
Answers will vary.

Equation	Your Partner's Scenario	Does the equation represent your partner's scenario?	Similarities between scenarios for the equation
		☐ Yes ☐ No	
		☐ Yes ☐ No	

1.6.4 COLLABORATION: WRITING REAL-WORLD SCENARIOS TO REPRESENT ALGEBRAIC EQUATIONS (CONTINUED)

TEACHER MOVES

Prompt students to take turns sharing their initial responses, similarities, and how their answers may have changed based on comparing responses with a partner.

DIFFERENTIATE INSTRUCTION

Consider providing select students with graphic organizers for common key words (i.e., split indicated division) to help students decode word problems.

QUESTIONING

For students who finish fast, have them switch partners and review 2 more equations/scenarios with a different partner.



1.6.5 Wrap-Up: Ticket Out the Door

1. Four equations are shown.

$$3x = -15$$

$$x - 3 = 15$$

$$\frac{x}{3} = 15$$

$$x + 3 = -15$$

Match an equation to each situation it represents. Then, complete the table. Not all equations will be used.

Situation	Equation	What does x represent?	Solution
a. On Tuesday, the temperature in Nunavut rose 3° from the previous day to -15° . What was the temperature on Monday?	$x + 3 = -15$	<i>The temperature on Monday</i>	$x = -18$
b. Tyanna is revamping her wardrobe. She decides to donate 3 old t-shirts. Now, she has 15 t-shirts. How many t-shirts did she have originally?	$x - 3 = 15$	<i>The number of t-shirts Tyanna originally had</i>	$x = 18$
c. A shipment of oranges was split into 3 equally sized crates. Each crate weighs 15 pounds. How much did the original shipment weigh?	$\frac{x}{3} = 15$	<i>The weight of the original shipment</i>	$x = 45$

1.6.5 WRAP-UP: TICKET OUT THE DOOR

DIFFERENTIATE INSTRUCTION

Consider using student responses from prior self-reflections to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.